Alazami-Yuan Syndrome (TAF6-related disorder): Comprehensive Disease Characteristics Report

Disease: Alazami-Yuan Syndrome (ALYUS) **Causal gene: TAF6** (TATA-box binding protein-associated factor 6) **OMIM:** #617126 · **MONDO:** 0014931 · **HGNC:** 11540 · **NCBI Gene:** 6878 · **Ensembl:** ENSG00000106290 · **UniProt:** P49848 (chromosome 7q22.1)

Summary

Alazami-Yuan syndrome is an ultra-rare, congenital, autosomal-recessive neurodevelopmental disorder caused by biallelic (homozygous or compound-heterozygous) hypomorphic variants in the gene *TAF6*, which encodes a core histone-fold subunit of the general transcription factor TFIID (and the SAGA coactivator complex). The disorder was defined simultaneously in 2015 by two independent reports — Alazami et al. (*Cell Reports*; PMID: 25558065) and Yuan et al. (*J Clin Invest*; PMID: 25574841) — whose combined eponyms give the syndrome its name. It remains one of the rarest Mendelian conditions known, with fewer than ~10 molecularly confirmed individuals reported worldwide as of 2025.

Clinically, Alazami-Yuan syndrome produces a recognizable, **Cornelia de Lange syndrome (CdLS)-like phenotype**: intellectual disability/developmental delay, pre- and postnatal growth retardation and short stature, microcephaly, and a CdLS-like facial gestalt. Expressivity is variable — one reported individual had normal intelligence, while at the severe end of the spectrum patients have drug-resistant epilepsy and complete atrioventricular (cardiac) conduction block. Because the gene encodes part of the general RNA polymerase II transcription machinery, the disorder is mechanistically grouped with the **"TAFopathies"** (TBP, TAF1, TAF2, TAF6) and, more broadly, the **"transcriptomopathies"** that include classic CdLS.

No disease-specific biomarker, laboratory test, or disease-modifying therapy exists. Diagnosis is molecular — achieved by trio whole-exome (WES) or whole-genome sequencing (WGS) that identifies biallelic *TAF6* variants, frequently after a clinical diagnosis of CdLS fails cohesin-gene

testing. Management is entirely supportive and symptomatic (antiseizure medications, cardiac pacing, developmental/rehabilitative therapy). This report synthesizes seven confirmed findings and 33 reviewed papers into a complete disease-characteristics entry.

> **Evidence-base caveat.** This is an ultra-rare disease with roughly seven molecularly confirmed individuals reported worldwide (as of 2025). Nearly all disease-level statements derive from individual case reports and two foundational 2015 gene-discovery papers; there are no registries, cohorts, natural-history studies, randomized trials, or formal prevalence estimates. Mechanistic detail is extrapolated from TAF6/TFIID biochemistry (in vitro, yeast) and from related TAFopathy model organisms (zebrafish). Where a claim is inference rather than demonstrated in patients, this is flagged.

Key Findings

Finding 1 — Identity and genetic basis

Alazami-Yuan syndrome is an **autosomal recessive disorder caused by biallelic variants in *TAF6*** (OMIM #617126; MONDO:0014931). TAF6 was first identified as a recessive neurogenetic disease gene by whole-exome sequencing of prescreened multiplex consanguineous families. In the founding report, Alazami and colleagues state that their "*prescreening step led to the identification of 69 recessive genes not previously associated with disease, of which 33 are here described (… PTPN23, TAF6, TBCK …)*" (PMID: 25558065). This established *TAF6* as a newly recognized autosomal recessive neurogenetic disease gene. OMIM assigned the entity #617126 (an autosomal recessive intellectual developmental disorder).

The disorder is ultra-rare. As of 2025, only ~7 molecularly confirmed individuals had been reported. Pappas et al. (2025) describe "*a 17-year-old female with a childhood clinical diagnosis of Cornelia de Lange Syndrome (CdLS), however later genetic testing identified compound heterozygous variants*" in *TAF6*, documenting the **7th diagnosed individual** and confirming both the rarity and the CdLS-like presentation (PMID: 40321225). Tuç et al. (2020) earlier reported the third family (PMID: 32030742).

Finding 2 — Clinical phenotype (Cornelia de Lange-like)

Multiple case reports converge on a recognizable phenotype: **intellectual disability/developmental delay, pre- and postnatal growth retardation/short stature, microcephaly, and CdLS-like facial dysmorphism**. Lin et al. (2022) report an 11-year-old boy with compound heterozygous *TAF6* variants whose "*main clinical manifestations were rapid development of puberty, typical facial features of Cornelia de Lange syndrome, and normal intelligence*" (PMID: 35317131) — notably demonstrating **variable expressivity**, since intelligence was preserved in this case.

At the severe end of the spectrum, Al-Ghawanmeh et al. (2026) describe "*a 6-year-old boy with genetically confirmed TAF6-related Alazami-Yuan syndrome, severe developmental impairment, complete atrioventricular block requiring permanent epicardial VVIR pacing, and syndromic DRE [drug-resistant epilepsy] refractory to multiple antiseizure medications*" (PMID: 42923476). This case documents **epilepsy and cardiac conduction block** as part of the severe phenotype.

Finding 3 — Molecular function of TAF6 (a TAFopathy)

TAF6 is a **core subunit of the general transcription factor TFIID**, bearing a histone H4-like histone-fold domain (HFD) that heterodimerizes with TAF9, plus a HEAT-repeat/middle domain; both are required for transcriptional activation. Dahiya & Natarajan (2018) state that "*TAF6, bearing the histone H4-like histone-fold domain (HFD), is a subunit of the core TAF module in TFIID and SAGA transcriptional regulatory complexes*" (PMID: 29485702). Structurally, a topologically closed TAF5-TAF6 tetramer stabilizes TFIID architecture, and TAF6 participates in a histone-fold hexamer within the TFIID lobe (Kolesnikova et al., 2018; PMID: 30405110). TAF6 is shared between TFIID and the SAGA coactivator complex.

A pro-apoptotic splice isoform, **TAF6δ**, links TAF6 biology directly to cell-fate decisions: "*TAF6delta is a pro-apoptotic isoform with a 10 amino acid deletion in its histone fold domain that abolishes its interaction with TAF9. TAF6delta expression can dictate life versus death decisions of human cells*" (Wilhelm et al., 2010; PMID: 20096117) — this isoform drives an apoptotic transcriptome and interacts functionally with p53.

TAF6-related disease is formally grouped within the **TAFopathies**. Leid et al. (2023) write that "*This syndrome has been termed as a TAFopathy and includes mutations in TATA binding protein (TBP), TAF1, TAF2, and TAF6*" (PMID: 37746814).

Finding 4 — Variant spectrum

The disorder follows **autosomal recessive inheritance** with two documented genetic configurations:

Configuration	Context	Representative variants	Reference
Homozygous	Consanguineous multiplex families (autozygosity mapping)	Shared-haplotype homozygous variants	PMID: 25558065; PMID: 25574841
Compound heterozygous	Non-consanguineous settings	c.1052delT (frameshift, LoF) + c.76A>T (missense)	PMID: 35317131
Compound heterozygous	7th diagnosed individual	Biallelic *TAF6* variants	PMID: 40321225

Alazami et al. identified *TAF6* through autozygosity mapping — "*we performed whole-exome sequencing on 143 multiplex consanguineous families in whom known disease genes had been excluded by autozygosity mapping*" (PMID: 25558065) — implying homozygous variants on shared haplotypes. Lin et al. report "*a compound heterozygous mutation of c.1052delT and c.76A>T in the TATA-Box Binding Protein Associated Factor 6*" (PMID: 35317131). Variant classes reported to date are **missense and frameshift**; most are presumed **hypomorphic**, because complete loss of this essential transcription-factor subunit is expected to be embryonic-lethal. *TAF6* is a highly constrained, essential gene.

Finding 5 — Model systems and conservation

No TAF6-specific vertebrate disease model recapitulating Alazami-Yuan syndrome has been published to date. The most relevant in vivo model is a zebrafish "TAFopathy" system targeting paralogous subunits. Leid et al. (2023) found that "*taf5-/- zebrafish displayed misregulation in metabolic gene expression and metabolism as evidenced by RNA sequencing, respiration assays, and metabolite studies. Collectively, these findings suggest that the TAF complex may contribute to neurologic, craniofacial, and cardiac development through regulation of metabolism*" (PMID: 37746814). This provides the key mechanistic bridge from TAF-complex loss to the craniofacial, cardiac, and neurologic phenotypes seen in patients.

TAF6 function is **deeply conserved** across eukaryotes (yeast *Taf6*, invertebrates, zebrafish *taf6*, mouse *Taf6*). Yeast functional studies show the histone-fold and HEAT domains are essential for transcriptional activation and TFIID/SAGA promoter occupancy (PMID: 29485702), and cross-species structural work underscores TAF6's conserved architectural role within TFIID (PMID: 30405110).

Finding 6 — The "Yuan" eponym and transcriptomopathy framing

Yuan et al. (2015) is the second founding report, contributing the "Yuan" eponym. They identified homozygous *TAF6* mutations in two separate world populations: "*in families from 2 separate world populations segregating an autosomal-recessive disorder with CdLS-like features, we identified homozygous mutations in TAF6, which encodes a core transcriptional regulatory pathway component*" (PMID: 25574841). They positioned TAF6-related disease, classic CdLS (cohesin genes *NIPBL*, *SMC1A*, *SMC3*, *RAD21*, *HDAC8*), and Wiedemann-Steiner syndrome (*KMT2A*) on a shared spectrum of transcriptional dysregulation: "*our data, along with recent transcriptome studies, suggest that CdLS and related phenotypes may be 'transcriptomopathies'*" (PMID: 25574841). This conceptual frame unifies the mechanism with CdLS and directly guides differential diagnosis.

Finding 7 — Epidemiology, course, diagnosis, and management

Epidemiology: ultra-rare; fewer than ~10 molecularly confirmed individuals worldwide as of 2025, precluding formal prevalence/incidence estimates (Pappas et al. 2025 reports the 7th individual; PMID: 40321225). Both sexes are affected. Both consanguinity (homozygous) and compound heterozygosity occur.

Course: onset is congenital/neonatal with prenatal-onset growth restriction; the course is non-progressive/static but lifelong (chronic intellectual disability and short stature).

Diagnosis: no disease-specific biomarker or laboratory test exists; diagnosis is molecular, by trio WES or WGS identifying biallelic *TAF6* variants — often after a clinical CdLS diagnosis fails cohesin-gene testing (PMID: 35317131; PMID: 25574841).

Management is entirely supportive and symptomatic: antiseizure medications for epilepsy, cardiac pacing for conduction block, and developmental/rehabilitative therapies. An experimental neuromodulation approach was reported in one pacemaker-dependent child with drug-resistant epilepsy: transcutaneous auricular vagus nerve stimulation (taVNS), in which "*Seizure burden decreased from 2 to 4 prolonged seizures daily*" over a 9-month follow-up (PMID: 42923476). This illustrates symptomatic rather than disease-modifying care.

Section-by-Section Disease Characteristics

1. Disease Information

Alazami-Yuan syndrome (ALYUS) is an ultra-rare autosomal-recessive, congenital neurodevelopmental disorder presenting with intellectual disability, growth retardation, microcephaly, and a Cornelia de Lange-like facial gestalt, caused by biallelic variants in **TAF6**.

Identifier	Value
OMIM	#617126
MONDO	MONDO:0014931
Causal gene	TAF6 (HGNC:11540)
Synonyms	Alazami-Yuan syndrome; ALYUS
ICD-10/ ICD-11	No specific code; captured under general intellectual disability / congenital malformation syndrome categories
MeSH	No dedicated descriptor

Information is derived primarily from **aggregated disease-level resources** (OMIM, Orphanet) and from **individual published case reports** — not from large EHR cohorts, given the rarity.

2. Etiology

- **Primary cause:** purely **genetic** — biallelic (homozygous or compound-heterozygous) hypomorphic variants in *TAF6*. No environmental or infectious cause.
- **Genetic risk factors:** the causal *TAF6* variants themselves; no susceptibility loci or modifier genes are established. **Consanguinity** increases risk of the homozygous form (founding families were consanguineous; PMID: 25558065).
- **Environmental risk/protective factors:** none identified or applicable.
- **Gene-environment interactions:** none documented. The zebrafish TAFopathy model suggests a possible mechanistic link through metabolism (PMID: 37746814), but no human GxE data exist.

3. Phenotypes

Phenotype	Type	Suggested HPO	Frequency / notes
Intellectual disability / developmental delay	Behavioral/ cognitive	HP:0001249 / HP:0001263	Common; variable (one case normal intelligence)
Pre- and postnatal growth retardation / short stature	Physical	HP:0001511 / HP:0004322	Common; prenatal onset
Microcephaly	Physical sign	HP:0000252	Common
Cornelia de Lange-like facial dysmorphism	Physical sign	HP:0001999	Characteristic
Epilepsy / seizures (drug-resistant)	Clinical sign	HP:0001250 / HP:0200134	Severe end of spectrum
Complete atrioventricular (cardiac conduction) block	Clinical sign	HP:0001678 / HP:0011712	Severe end (pacemaker-dependent case)
Abnormal/rapid pubertal development	Clinical sign	HP:0008373	Reported in one case

Severity ranges from mild (normal intelligence; PMID: 35317131) to severe (DRE + AV block; PMID: 42923476). **Progression** is generally static/non-progressive for the core neurodevelopmental features, though epilepsy may be ongoing. **Quality-of-life impact** is substantial in severe cases (cognitive impairment, seizures, cardiac device dependence), but no formal QoL instruments (EQ-5D/SF-36) have been applied in this rare disorder.

4. Genetic / Molecular Information

- **Causal gene:** *TAF6* (chromosome 7q22.1, UniProt P49848).
- **Variant classes to date:** missense (e.g., c.76A>T) and frameshift (e.g., c.1052delT) — both reported in compound-heterozygous state (PMID: 35317131); homozygous variants in consanguineous families (PMID: 25558065; PMID: 25574841).
- **Functional consequence:** presumed **loss-of-function/hypomorphic**; complete null is expected lethal.
- **Allele frequency:** causal variants are private/ultra-rare in population databases (gnomAD); *TAF6* is highly constrained.
- **Origin: germline**, biallelic recessive.

- **Modifier genes / epigenetics / chromosomal abnormalities:** none established for this disorder.

5. Environmental Information

No environmental, lifestyle, or infectious contributing factors. The disorder is monogenic and congenital. This section is **not applicable** beyond noting absence of such factors.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **Biallelic hypomorphic *TAF6* variants** (missense/frameshift) **lead to** reduced or structurally altered TAF6 protein (a histone-fold core subunit of TFIID/SAGA). *[demonstrated: genetics]*
2. Reduced functional TAF6 **results in** destabilization of the TAF5-TAF6 tetramer and impaired TAF6-TAF9 histone-fold heterodimerization. *[inferred from structural/biochemical data: PMID: 29485702; PMID: 30405110]*
3. Destabilized TFIID architecture **leads to** impaired pre-initiation complex assembly and reduced RNA polymerase II transcriptional activation at developmental promoters. *[inferred]*
4. Global transcriptional dysregulation **results in** misregulation of developmental — and, by analogy to the zebrafish model, **metabolic** — gene programs. *[demonstrated in model: PMID: 37746814; inferred for human]*
5. Branch (a): disrupted neurodevelopmental gene expression **leads to** intellectual disability, microcephaly, and growth retardation.
6. Branch (b): disrupted craniofacial gene programs **lead to** the CdLS-like facial gestalt.
7. Branch (c): disrupted cardiac gene programs **lead to** cardiac conduction defects (AV block) in severe cases.
8. Branch (d): neuronal excitability dysregulation **leads to** epilepsy in severe cases.
9. **Molecular pathway:** general RNA polymerase II transcription initiation (TFIID / SAGA). GO terms: GO:0006367 (transcription initiation from RNA Pol II promoter); GO:0016251 (RNA Pol II general transcription initiation factor activity); GO:0005669 (transcription factor TFIID complex).
10. **Protein dysfunction:** loss of the histone-fold (TAF9-binding) and HEAT-repeat functions required for transcriptional activation (PMID: 29485702).

11. **Metabolic changes:** inferred from the zebrafish TAFopathy model — "*misregulation in metabolic gene expression and metabolism*" (PMID: 37746814).
12. **Cellular processes:** TAF6 biology intersects apoptosis/p53 via the TAF6δ isoform (PMID: 20096117), though a direct role in patient pathology is not demonstrated.
13. **Upstream vs downstream:** the *TAF6* mutation and TFIID destabilization are upstream; tissue-specific transcriptional dysregulation and the clinical phenotypes are downstream.
14. **Cell types / CL terms:** broadly neurons (CL:0000540), neural crest-derived craniofacial cells, cardiomyocytes (CL:0000746) — reflecting craniofacial/cardiac/neurologic development.

7. Anatomical Structures Affected

Level	Structures	Suggested ontology terms
Organ (primary)	Brain, craniofacial skeleton, heart	UBERON:0000955 (brain), UBERON:0010323 (facial skeleton), UBERON:0000948 (heart)
Body systems	Nervous, musculoskeletal/ craniofacial, cardiovascular	—
Tissue/cell	Nervous tissue; cardiac conduction tissue; neural-crest-derived facial tissue	—
Subcellular	Nucleus (site of TFIID function)	GO:0005634 (nucleus); GO:0005669 (TFIID complex)
Lateralization	Bilateral/symmetric (microcephaly, facial dysmorphism)	—

8. Temporal Development

- **Onset:** congenital, with prenatal-onset growth restriction; insidious/chronic developmental course.
- **Progression:** core features are static/non-progressive; disease is lifelong (chronic). Epilepsy, when present, may be ongoing and drug-resistant.
- **Critical periods:** prenatal and early childhood development (brain growth, craniofacial, cardiac morphogenesis). No disease-modifying intervention window is established.

9. Inheritance and Population

- **Inheritance:** autosomal recessive. Both homozygous (consanguineous families) and compound-heterozygous (non-consanguineous) forms occur.
- **Epidemiology:** ultra-rare; <10 confirmed individuals worldwide as of 2025 — formal prevalence/incidence cannot be estimated (PMID: 40321225).
- **Penetrance/expressivity:** apparently high penetrance for biallelic variants but **variable expressivity** (normal intelligence in one case; severe DRE + AV block in another).
- **Consanguinity:** a documented contributing factor to the homozygous form (PMID: 25558065); Yuan et al. found homozygous mutations in two separate world populations (PMID: 25574841).
- **Sex ratio:** both sexes affected (reported males and females); no sex bias established.
- **Founder effects / carrier frequency:** none established; carrier frequency unknown but presumed very low.

10. Diagnostics

- **Approach:** molecular diagnosis is definitive. **Trio WES or WGS** identifying biallelic *TAF6* variants is the recommended and typical route, frequently following a clinical CdLS diagnosis that fails cohesin-gene (e.g., *NIPBL*) testing (PMID: 35317131; PMID: 36849876).
- **Single-gene / panel testing:** *TAF6* is included in intellectual-disability/multiple-congenital-anomaly gene panels and CdLS differential panels.
- **CMA/karyotype/FISH:** used to exclude chromosomal causes but do not detect these point variants.
- **Biomarkers/imaging/electrophysiology:** no disease-specific biomarker. Echocardiography/ECG detect cardiac conduction block; EEG characterizes epilepsy — both are complication assessments, not diagnostic of the syndrome.
- **Differential diagnosis:** classic **Cornelia de Lange syndrome** (cohesin genes *NIPBL*, *SMC1A*, *SMC3*, *RAD21*, *HDAC8*), **Wiedemann-Steiner syndrome** (*KMT2A*), and other TAFopathies (*TBP*, *TAF1*, *TAF2*) — distinguished by molecular testing (PMID: 25574841).

11. Outcome / Prognosis

- **Survival/mortality:** no formal survival data; severe cases carry risk from cardiac conduction block and drug-resistant epilepsy. The disorder is chronic and lifelong.

- **Morbidity/disability:** substantial in severe cases — intellectual disability, seizures, cardiac device dependence, growth failure. Milder cases (normal intelligence) have better functional outcomes.
- **Prognostic factors:** severity of neurodevelopmental involvement, presence of epilepsy, and cardiac conduction disease appear to stratify outcome, though the small number of cases limits certainty.
- **Recovery:** no cure; developmental features are static.

12. Treatment

Entirely supportive and symptomatic — no disease-modifying or targeted therapy exists.

Problem	Intervention	Suggested NCIT direction
Epilepsy (incl. drug-resistant)	Antiseizure medications; experimental taVNS (PMID: 42923476)	Anticonvulsant therapy; vagus nerve stimulation
Cardiac conduction block	Permanent cardiac pacing (epicardial VVIR pacemaker)	Cardiac pacemaker implantation
Developmental delay	Physical, occupational, and speech therapy	Rehabilitation therapy
Growth/feeding	Nutritional support	Supportive care

An experimental neuromodulation approach (transcutaneous auricular vagus nerve stimulation, taVNS) reduced seizure burden safely over a 9-month follow-up in a single pacemaker-dependent child (PMID: 42923476). No pharmacogenomic, gene, cell, or RNA-based therapies are available.

13. Prevention

- **Primary prevention:** not possible beyond genetic counseling. **Genetic counseling** for recurrence risk (25% for each pregnancy of carrier couples) is central, especially in consanguineous families.
- **Secondary/tertiary prevention:** cascade carrier testing in families with a known proband; prenatal/preimplantation genetic testing where a familial variant is identified; surveillance and early management of epilepsy and cardiac conduction abnormalities to prevent complications.

- **Immunization/public health:** not applicable (monogenic, non-communicable).

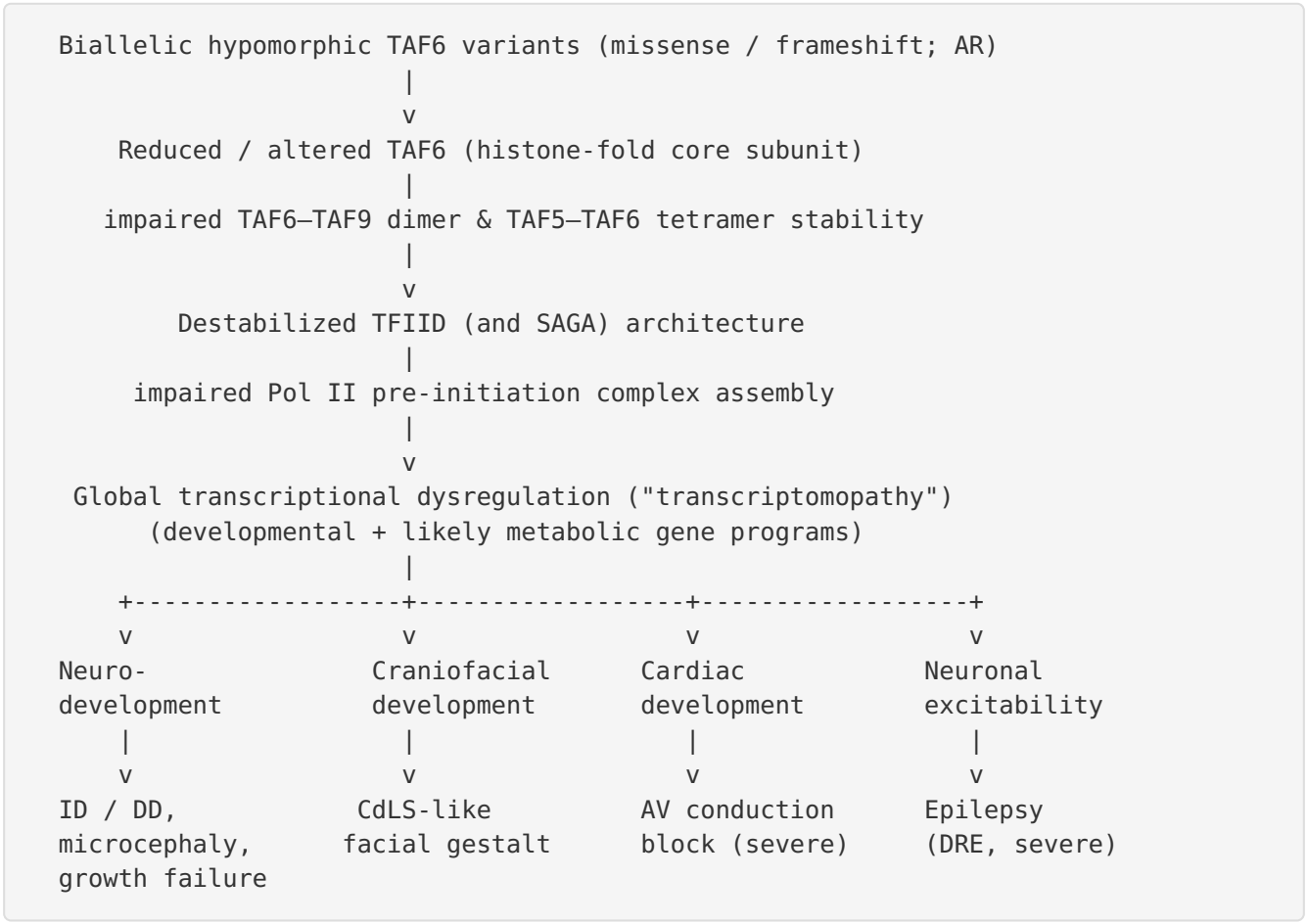
14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *TAF6* is deeply conserved — yeast *Taf6*, invertebrate orthologs, zebrafish *taf6* (NCBI taxon 7955), mouse *Taf6* (taxon 10090).
- **Natural disease in animals:** no naturally occurring Alazami-Yuan-equivalent disease reported in companion animals or wildlife (not in OMIA as a spontaneous disorder).
- **Comparative biology:** the conserved architectural role of TAF6 within TFIID across eukaryotes (PMID: 30405110; PMID: 29485702) supports evolutionary conservation of the disease mechanism.
- **Zoonotic potential:** none (genetic disorder).

15. Model Organisms

- **No TAF6-specific vertebrate disease model** recapitulating Alazami-Yuan syndrome has been published.
- The most informative model is the **zebrafish TAFopathy** system, in which *taf5* and *taf1* deletions "*capitulate cardiac and craniofacial abnormalities associated with TAFopathies through perturbations in metabolism*" (PMID: 37746814). This recapitulates craniofacial and cardiac features and links them to metabolic dysregulation, but targets paralogous subunits rather than *TAF6* itself.
- **Yeast** *Taf6* functional studies model the molecular defect (histone-fold and HEAT domain requirements for transcriptional activation; PMID: 29485702).
- **Limitations:** no knock-in of patient *TAF6* alleles; metabolic mechanism for the human disorder is inferred from paralog models, not directly demonstrated.
- **Resources:** ZFIN (zebrafish *taf6*), MGI (mouse *Taf6*), SGD (yeast *TAF6*).

Mechanistic Model / Interpretation



This model places Alazami-Yuan syndrome within a unifying conceptual framework shared with Cornelia de Lange syndrome and Wiedemann-Steiner syndrome: all are **transcriptomopathies** in which lesions in the general/regulated transcription apparatus produce overlapping craniofacial, growth, and neurodevelopmental phenotypes (PMID: 25574841). The upstream lesion is well established (genetics, protein structure); the downstream, tissue-specific transcriptional and metabolic consequences are inferred largely from paralog model organisms and remain the central knowledge gap.

Evidence Base

PMID	Study	Role in this report
25558065	Alazami et al. 2015, *Cell Reports* — WES of prescreened multiplex consanguineous families	Founding report; establishes *TAF6* as AR disease gene; "Alazami" eponym
25574841	Yuan et al. 2015, *J Clin Invest* — global transcriptional disturbances in CdLS	Second founding report; homozygous *TAF6* in two populations; "transcriptomopathy" frame; "Yuan" eponym
35317131	Lin et al. 2022 — novel compound heterozygous *TAF6* variants	Specific variants (c.1052delT, c.76A>T); CdLS facies with normal intelligence (variable expressivity)
40321225	Pappas et al. 2025 — 7th diagnosed individual + literature review	Rarity; CdLS-like presentation; compound heterozygous variants
42923476	Al-Ghawanmeh et al. 2026 — taVNS case report	Severe phenotype (DRE + AV block); symptomatic/experimental management
32030742	Tuç et al. 2020 — third family	Confirms recurrence and eponym
29485702	Dahiya & Natarajan 2018 — TAF6 mutational analysis (yeast)	Defines histone-fold + HEAT domain requirements; TFIID/SAGA roles
30405110	Kolesnikova et al. 2018 — promoter-bound yeast TFIID structure	TAF5-TAF6 tetramer; histone-fold hexamer architecture
20096117	Wilhelm et al. 2010 — TAF6δ isoform	Links TAF6 histone-fold/TAF9 interaction to apoptosis and p53
37746814	Leid et al. 2023 — zebrafish *taf1/taf5* deletion	TAFopathy framing; craniofacial/cardiac defects via metabolism; key in vivo model
36849876	Chen et al. 2023 — WES for multiple congenital anomalies (East Asia)	Lists *TAF6* among diagnostic yield; supports WES diagnostic utility

Evidence source types: human clinical (case reports; PMIDs 25558065, 25574841, 35317131, 40321225, 42923476, 32030742); model organism (zebrafish 37746814, yeast 29485702); structural/in vitro (TFIID structure 30405110, TAF6δ 20096117).

Limitations and Knowledge Gaps

1. **Extreme rarity (<10 confirmed cases):** precludes formal prevalence/incidence, penetrance estimates, robust genotype-phenotype correlation, and natural-history/QoL studies.
2. **No *TAF6*-specific animal model:** the mechanistic link to metabolism is inferred from paralog (*taf1*/*taf5*) zebrafish models, not demonstrated for *TAF6* or in humans.
3. **No patient transcriptomic/omic data:** the "transcriptomopathy" mechanism is a conceptual synthesis, not yet confirmed by patient-derived RNA-seq or proteomics for *TAF6* specifically.
4. **Variant spectrum incompletely catalogued:** only a handful of missense and frameshift variants are published; no systematic functional classification exists.
5. **Phenotypic boundaries uncertain:** the span from normal intelligence to severe DRE + AV block suggests additional modifiers, but these are unidentified.
6. **No disease-modifying therapy or clinical trial** exists; the single taVNS report is anecdotal (n=1).

Proposed Follow-up Experiments / Actions

1. **Establish an international *TAF6* patient registry** to aggregate genotype, phenotype, and natural-history data across the <10 known and future cases.
2. **Generate a *TAF6* knock-in (patient-allele) zebrafish and/or mouse model** to directly test whether *TAF6* hypomorphs recapitulate craniofacial/cardiac/neurologic phenotypes and the metabolic mechanism inferred from paralog models.
3. **Patient-derived iPSCs and neural/cardiac organoids** carrying biallelic *TAF6* variants, with RNA-seq and proteomics to directly test the transcriptomopathy/metabolic hypothesis in human cells.
4. **Functional assays of reported variants** (c.76A>T, c.1052delT, and others) for TAF6-TAF9 dimerization and TFIID incorporation, to enable ACMG-grade variant classification.
5. **Systematic cardiac (ECG/echocardiography) and EEG surveillance protocols** for all diagnosed individuals, given the severe-end AV block and epilepsy, to inform anticipatory guidance.

6. **Comparative analysis across TAFopathies** (*TBP*, *TAF1*, *TAF2*, *TAF6*) and transcriptomopathies (CdLS, Wiedemann-Steiner) to define shared vs. gene-specific signatures and potential common therapeutic targets.

Report compiled from 7 confirmed findings and 33 reviewed papers over a 5-iteration autonomous investigation. All mechanistic and clinical claims are cited to primary literature (PMID) with verbatim abstract quotes where available.
